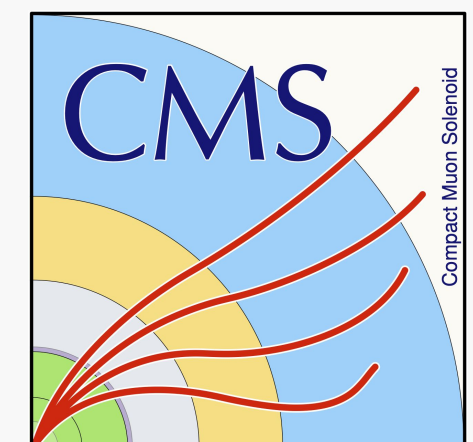
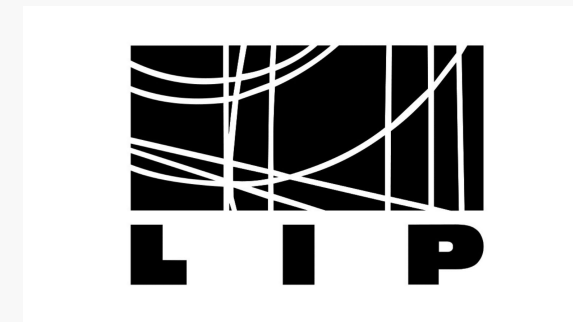


LIP-CMS weekly meeting

05/02/2025

Exploring new physics with photon interactions at the LHC
Madalena Ferreira

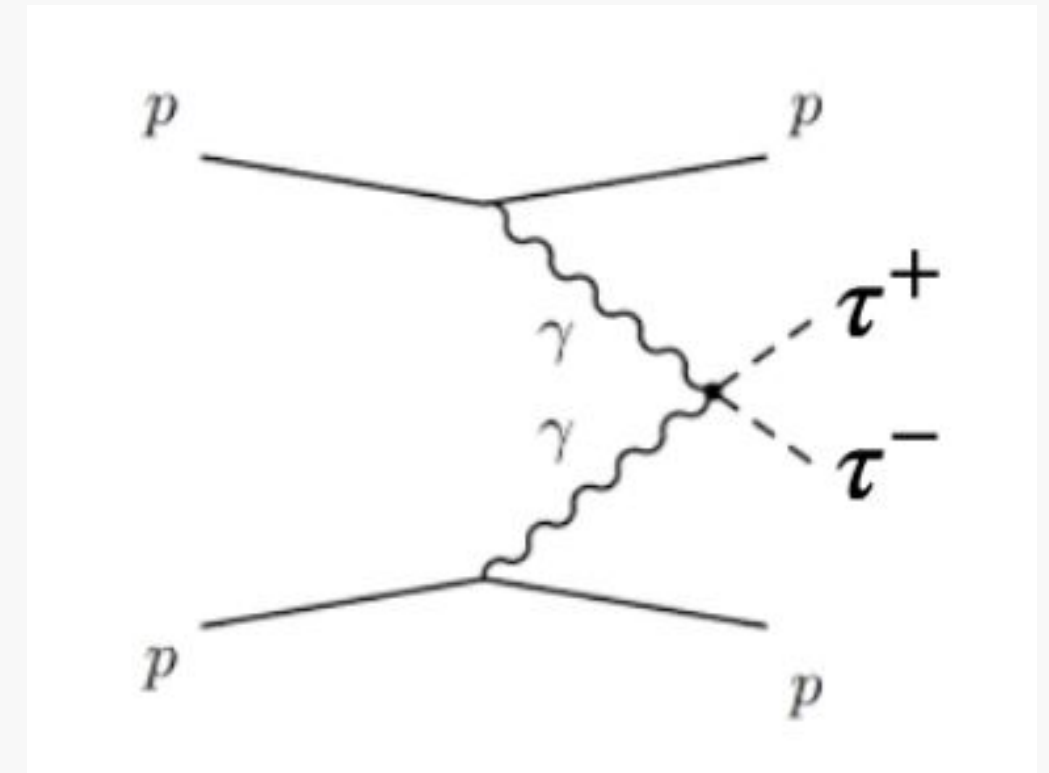


Tau anomalous magnetic moment



- Tau anomalous coupling refer to the deviations in the interactions involving tau leptons compared to the predictions made by the standard model;
- The tau anomalous magnetic moment, a_τ measures the deviation of the tau's magnetic dipole moment from the value predicted by the Dirac equation ($g=2$).
- Such deviations could point to physics beyond the standard model;
- Central exclusive production of tau pairs ($\tau^+\tau^-$) has direct sensitivity to a_τ through photon-photon interactions ($\gamma\gamma\rightarrow\tau^+\tau^-$), and can be used to study the gyromagnetic factor (g) of the τ lepton;

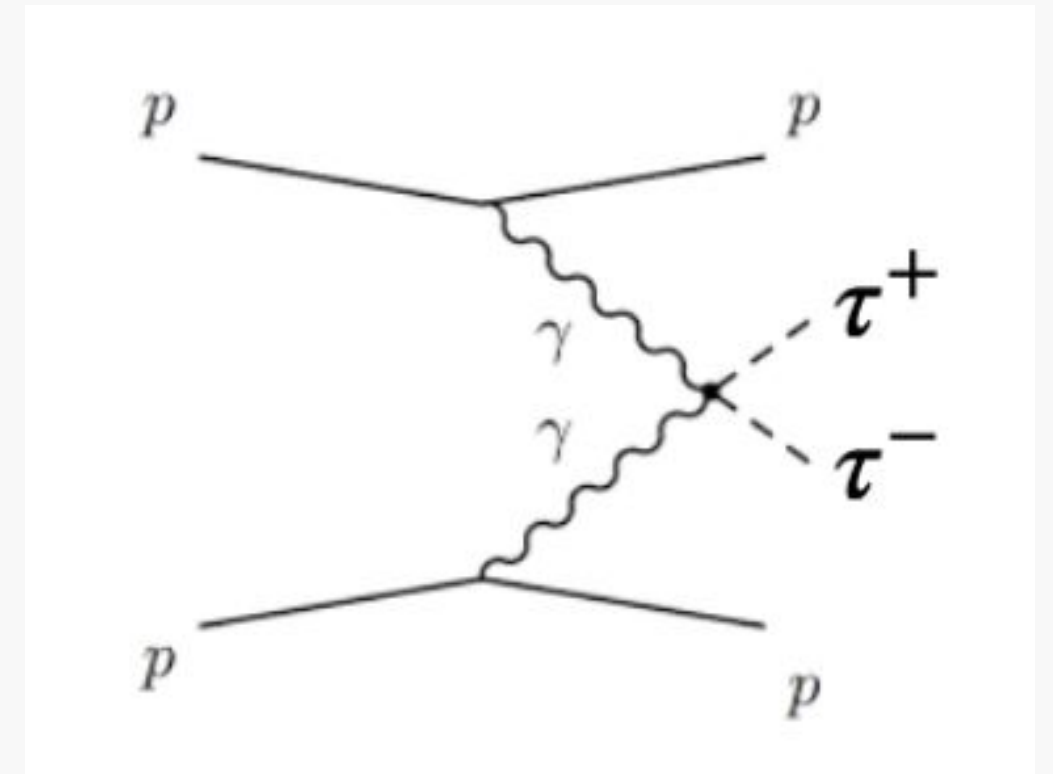
$$pp \rightarrow pX(\rightarrow \tau\tau)p,$$



Central exclusive production of tau pairs



- In this interaction protons are tagged by the by PPS, while the decay products of the X system are detected by the CMS central detector.
- Focus on the fully hadronic decay mode (both taus decay hadronically)
- Other decay modes
 - e + τ h decay mode: one τ decays hadronically and the other into an electron + two neutrinos;
 - μ + τ h decay mode: one τ decays hadronically and the other into a muon + two neutrinos;
 - e + μ decay mode: one τ decays in an electron + two neutrinos, and the other τ decays in a muon + two neutrinos, $\tau_e \tau_\mu$.



Existing analysis/results



- Recent studies of the muon magnetic moment show that it does not match theoretical predictions. If the tau magnetic moment also deviates from the standard model predictions, it could confirm new physics.
- Matteo's analysis studies the four mentioned decay modes using data from 2018 ($\sqrt{s}=13$ TeV, 54.9^{-1}).
 - Uses BDT to separate dominant backgrounds such as Drell-Yan, QCD and tt jets.
 - Sets an upper limit on the production cross section;
 - Sets constraints on the tau gyromagnetic factor.

Existing analysis/results

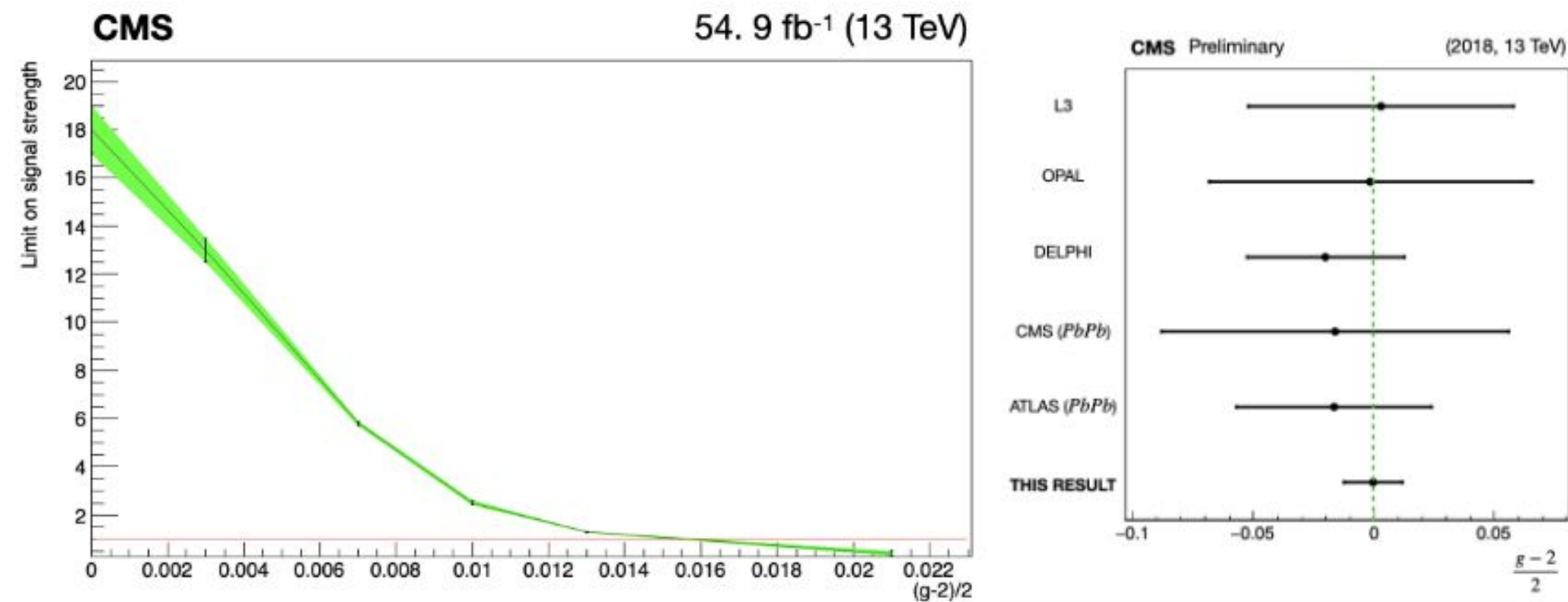


Figure 6: Limits on the signal strength as a function of a_τ (left); limits on a_τ for the different results (right).

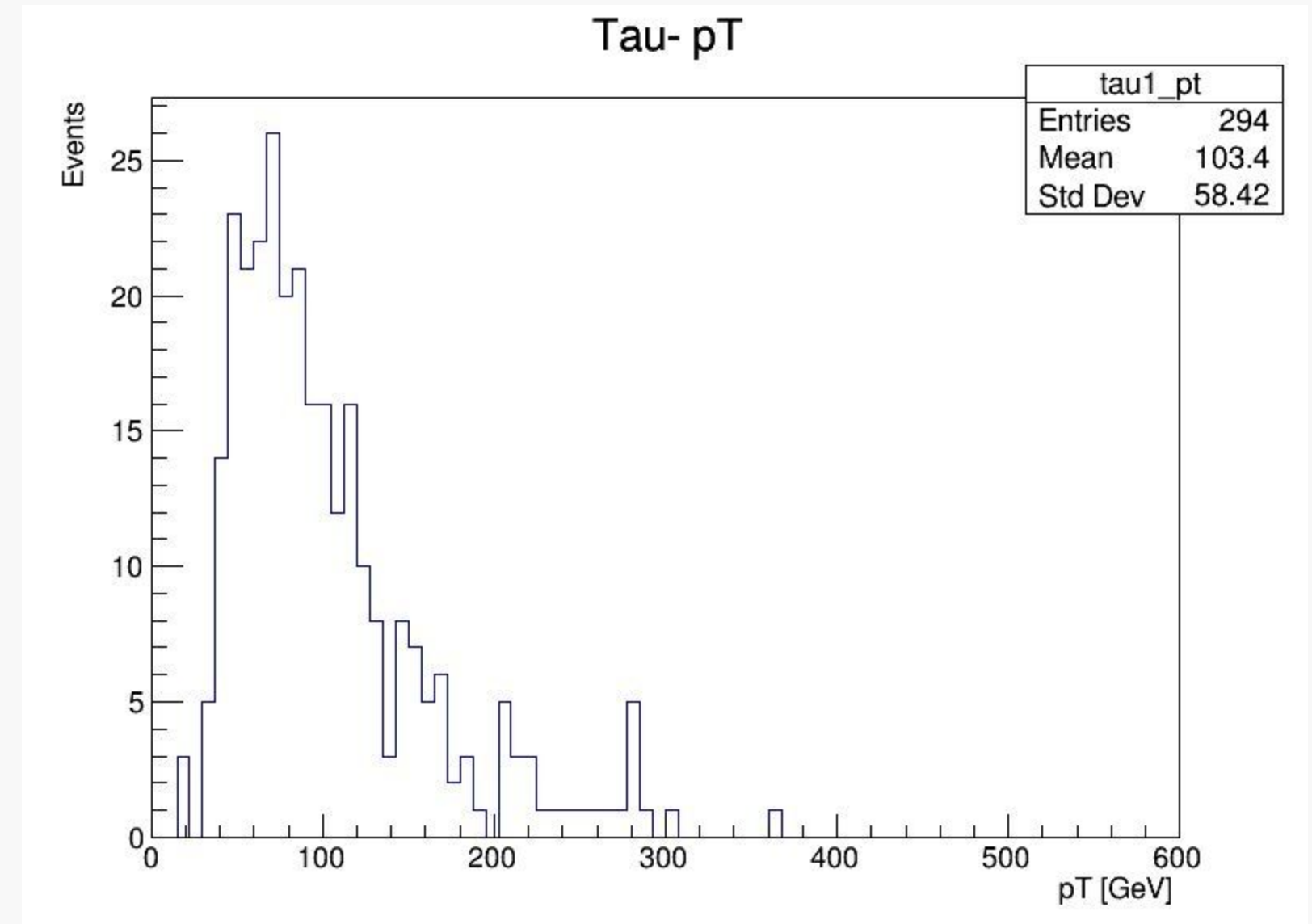
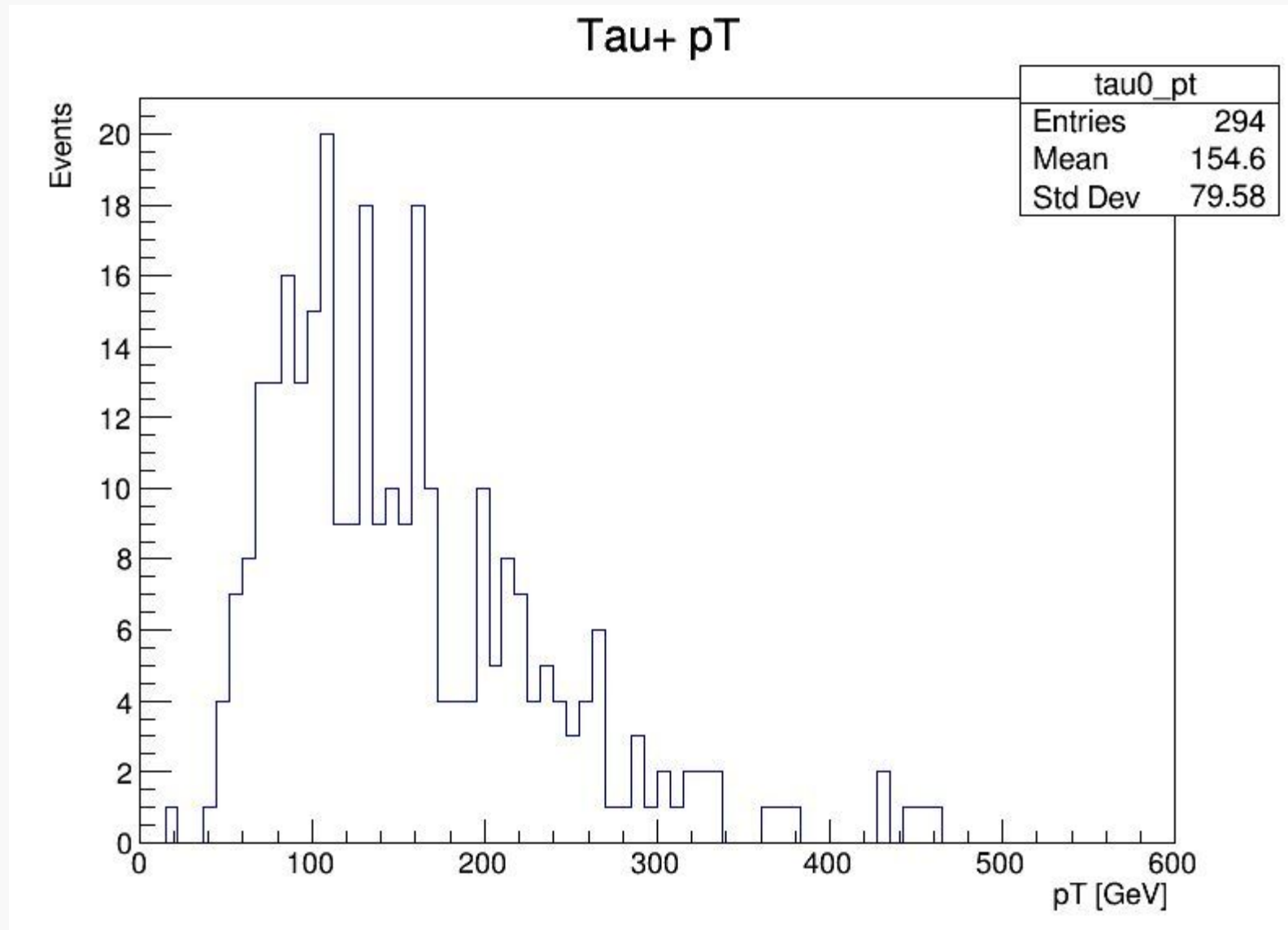
Variables of interest



- Transverse moment (pT) of the tau system
- System Rapidity: vectorial sum of the rapidity (Y) of the $\tau^+\tau^-$ pair decay products
- Acoplanarity: characterised by the presence of back-to-back τ s
- $M_{\text{CMS}} - M_{\text{PPS}}$
- $Y_{\text{CMS}} - Y_{\text{PPS}}$
- Proton energy loss, ξ
- Rapidity distributions of the tau system

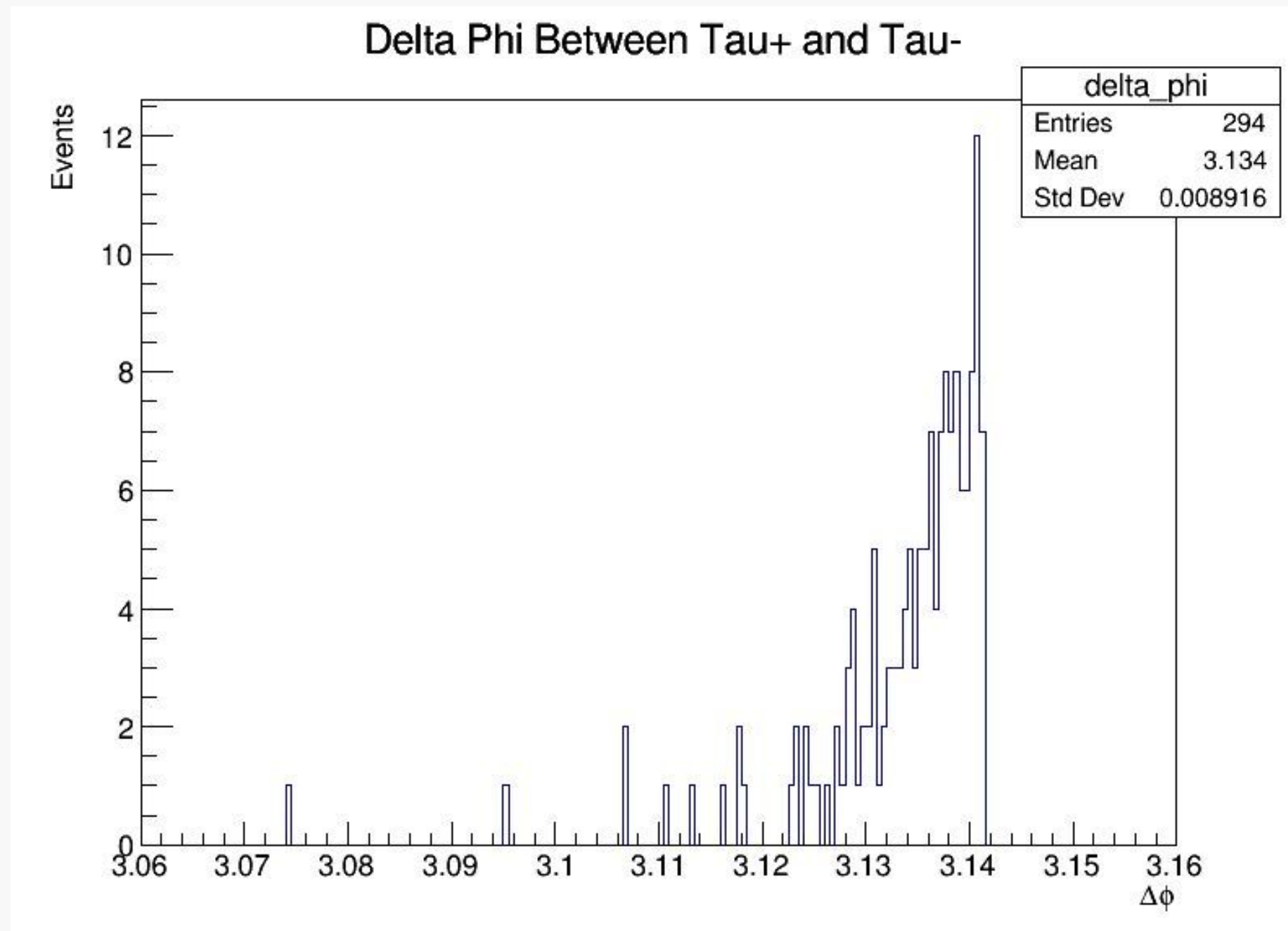
Plots

Monte carlo generated sample, 2018 run, 23k events



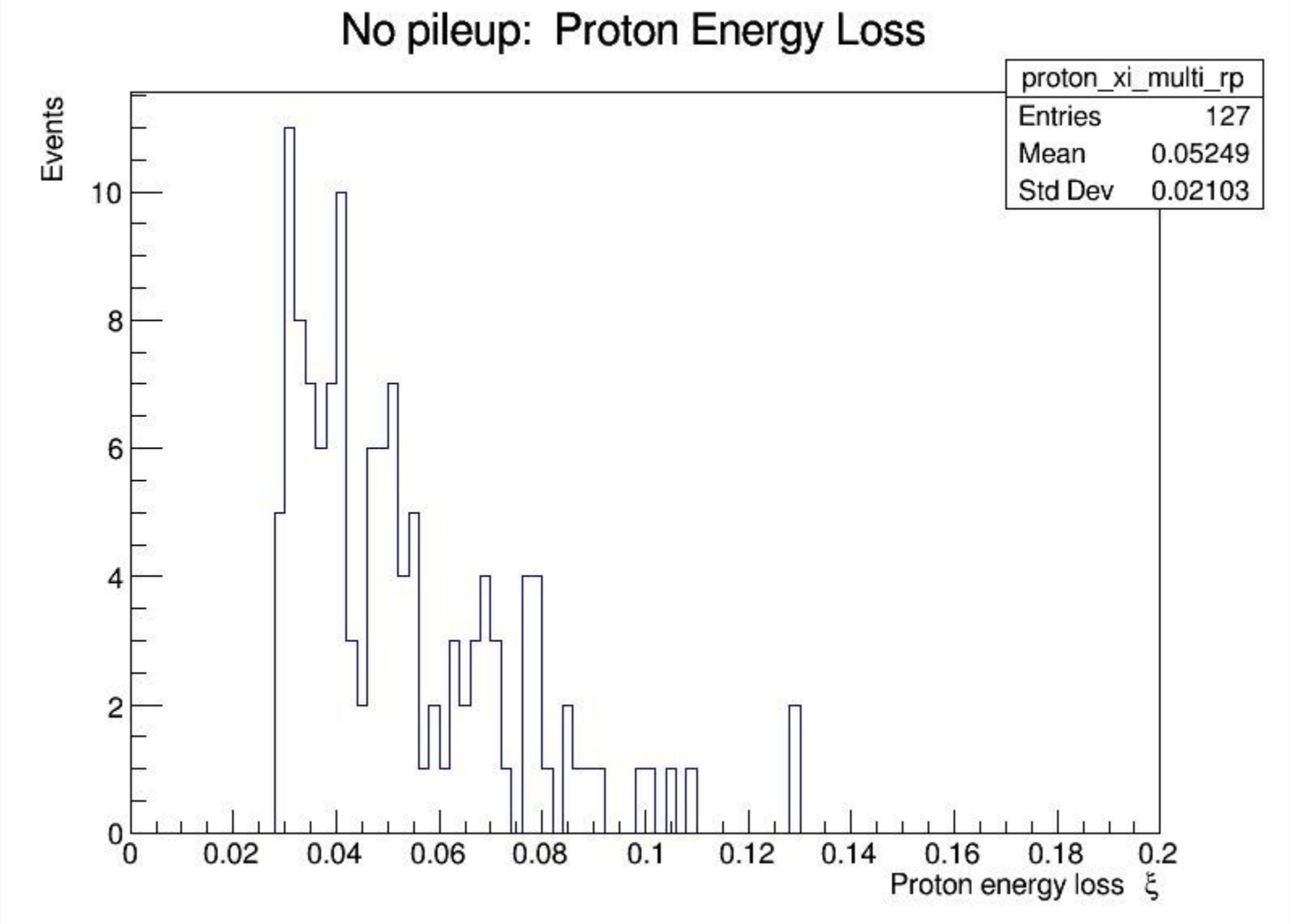
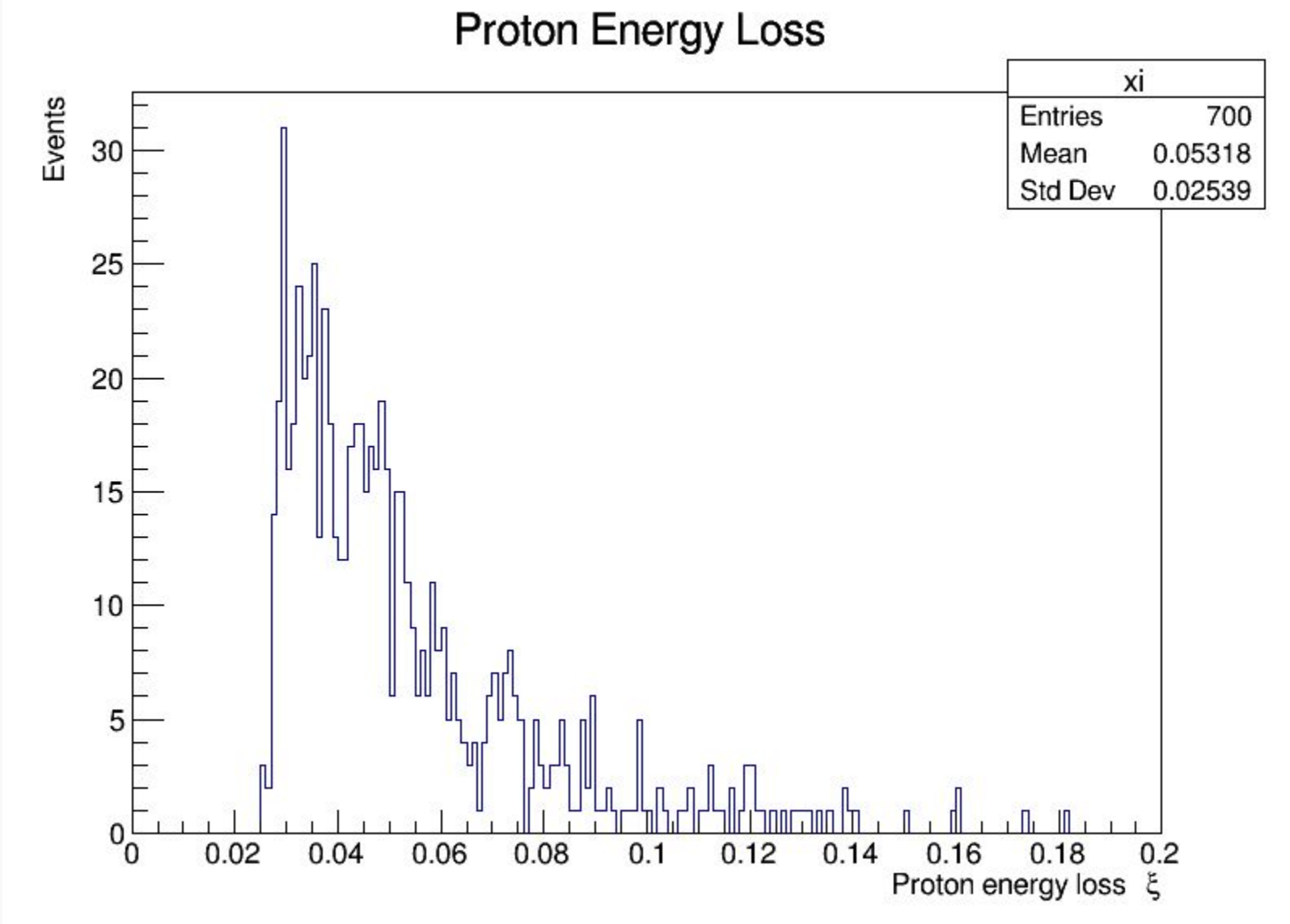
Transverse momentum: momentum of the tau-pair system projected onto the plane perpendicular to the beamline (z-axis)

Plots



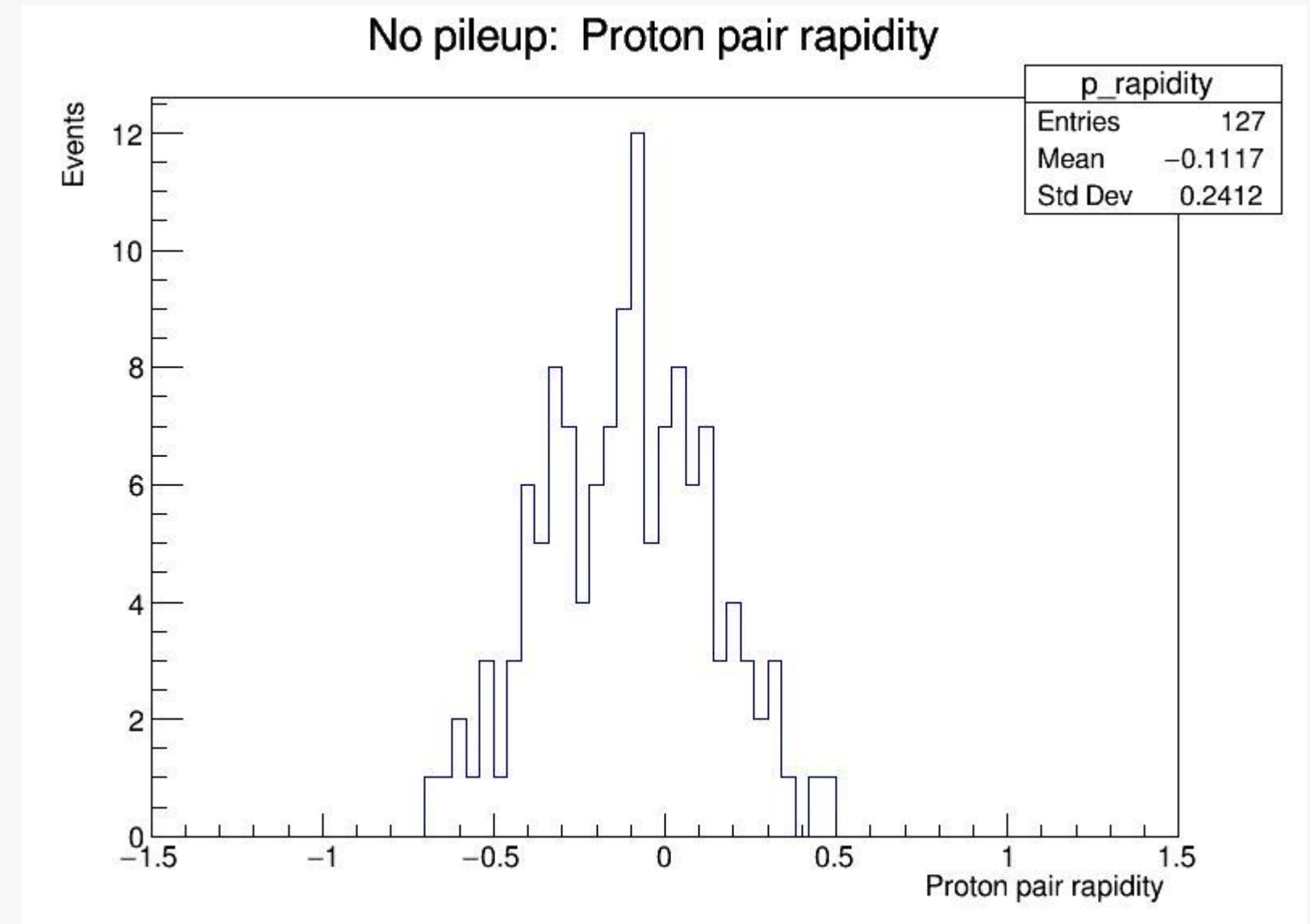
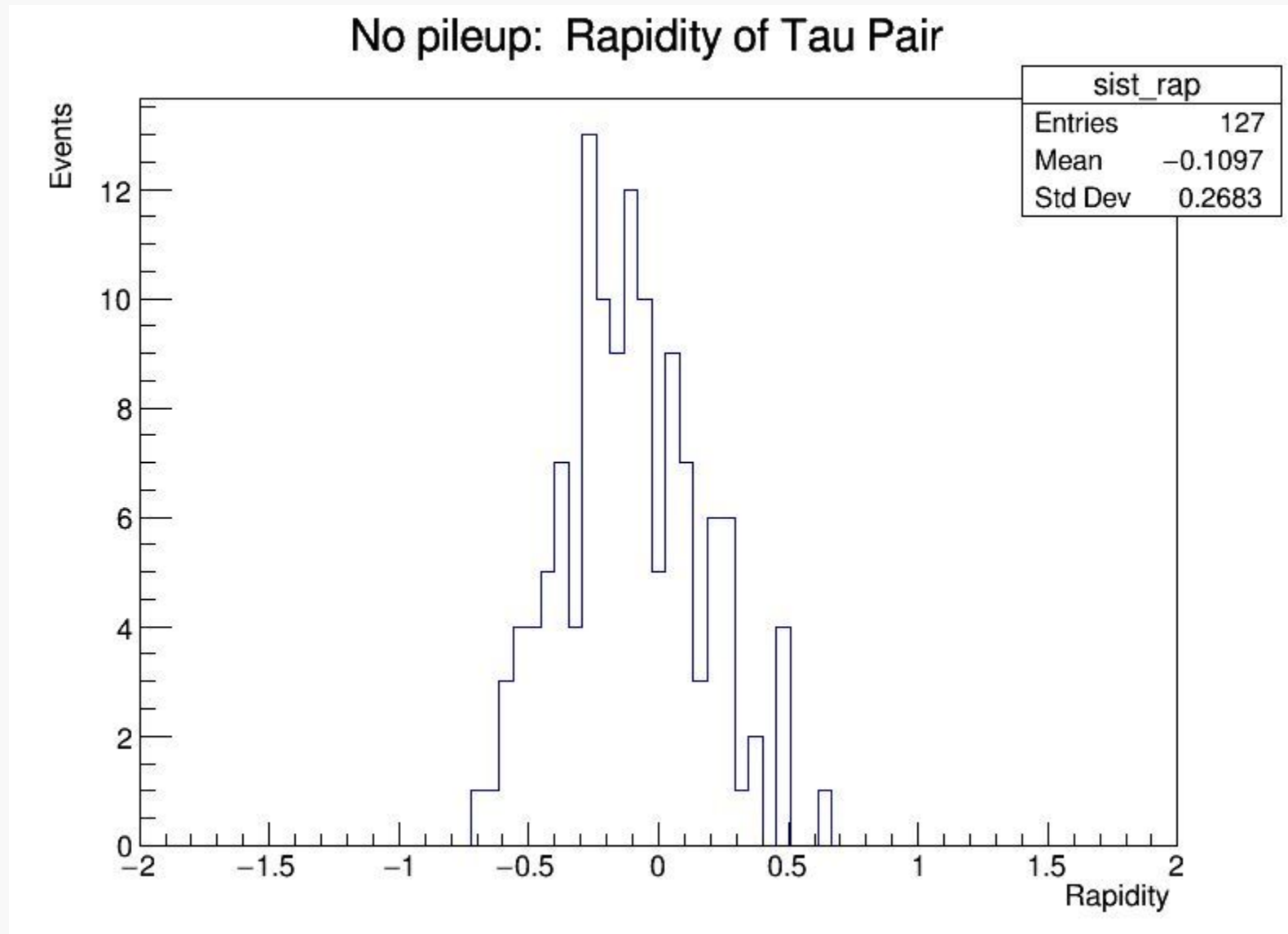
$$\Delta\phi = \phi(\tau_1) - \phi(\tau_0)$$

Plots



Fractional energy loss of protons tagged by PPS

Plots

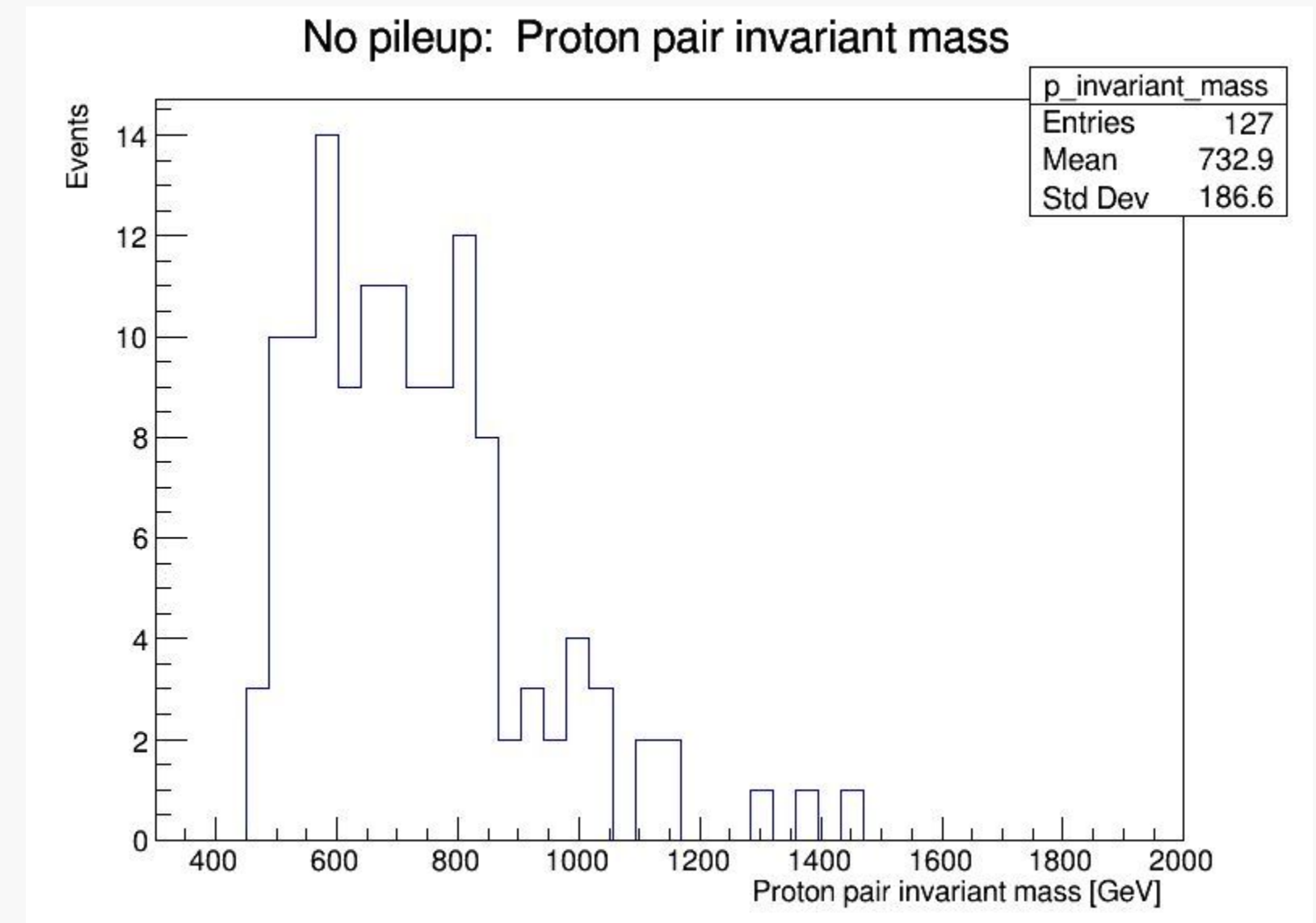
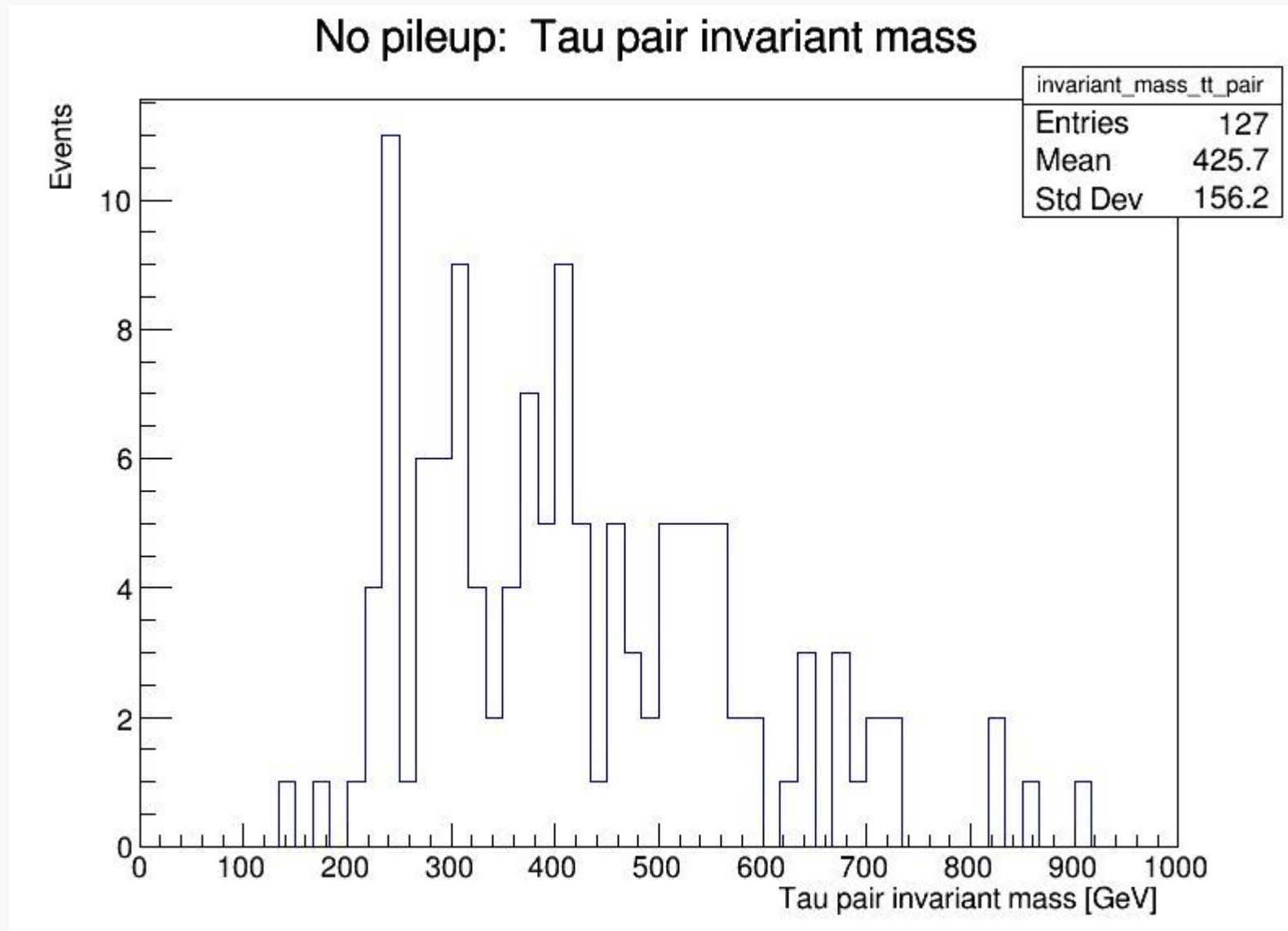


$$Y_X = \frac{1}{2} \left(\frac{E + p_z}{E - p_z} \right)$$

$$Y_X = \frac{1}{2} \log(\xi_1/\xi_2)$$

- E is total energy of tau pair
- P_z is total longitudinal moment of tau pair system

Plots

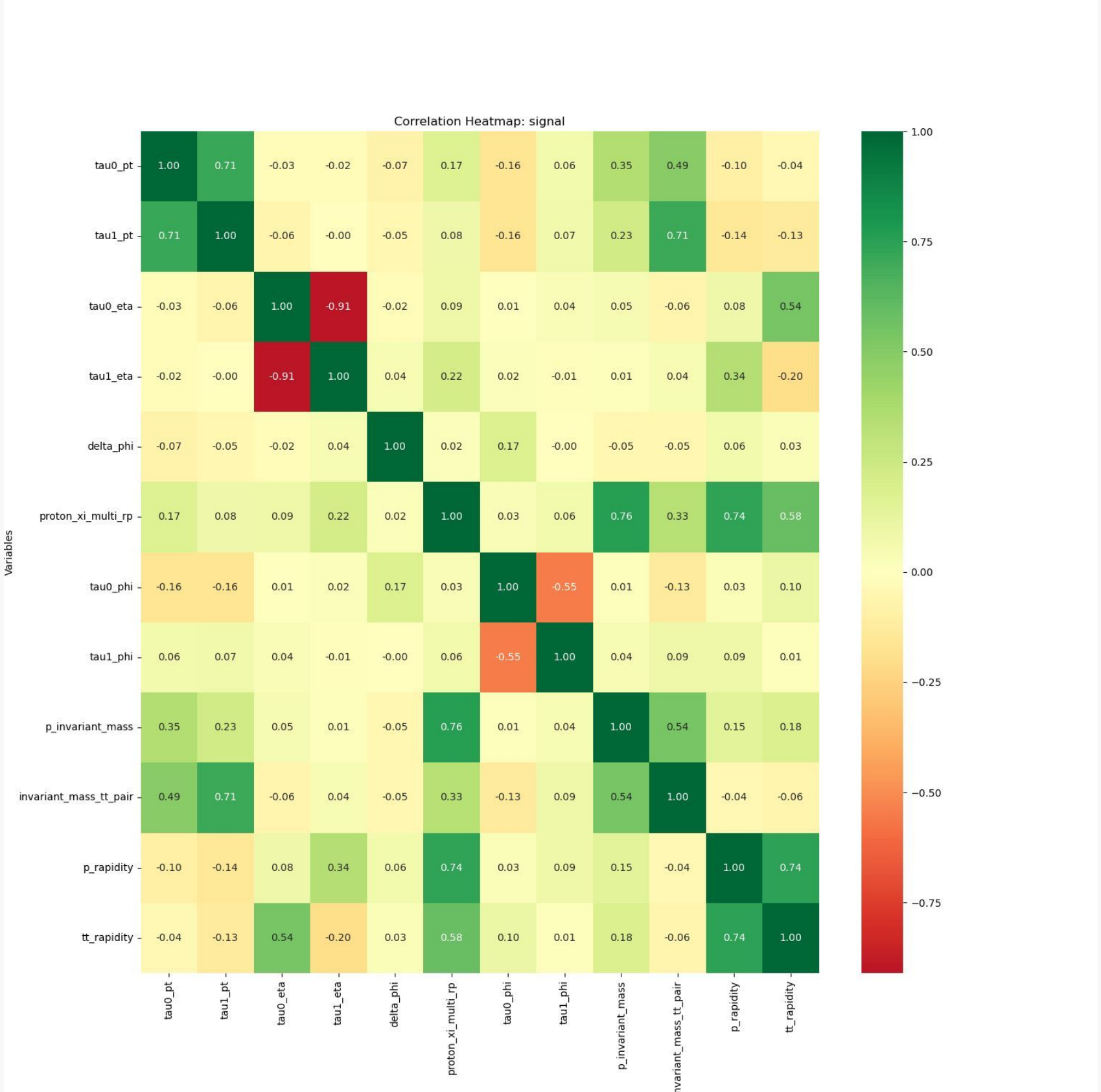


Used: $M^2 = 2pT_1pT_2(\cosh(\Delta\eta) - \cos(\Delta\phi))$

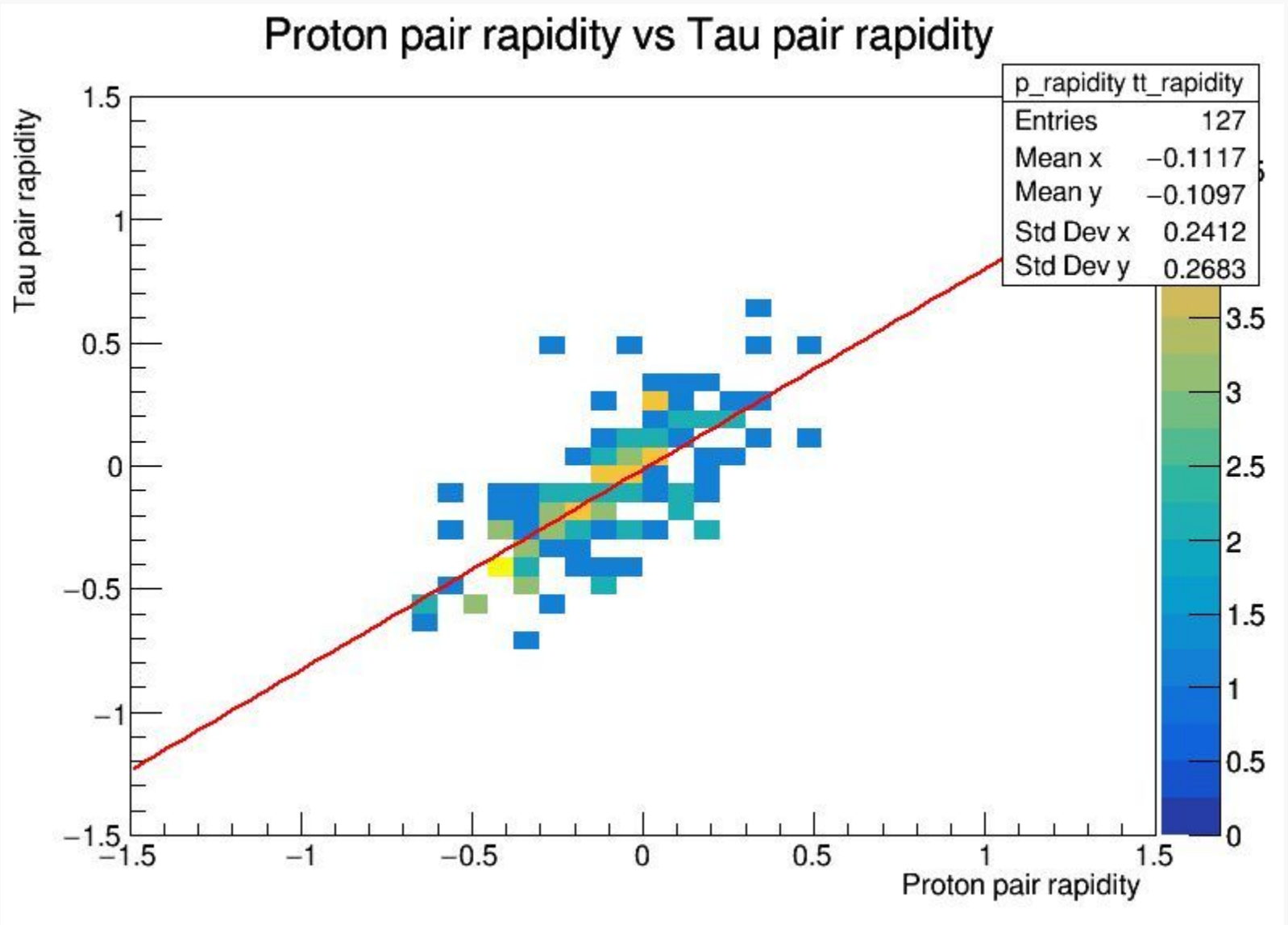
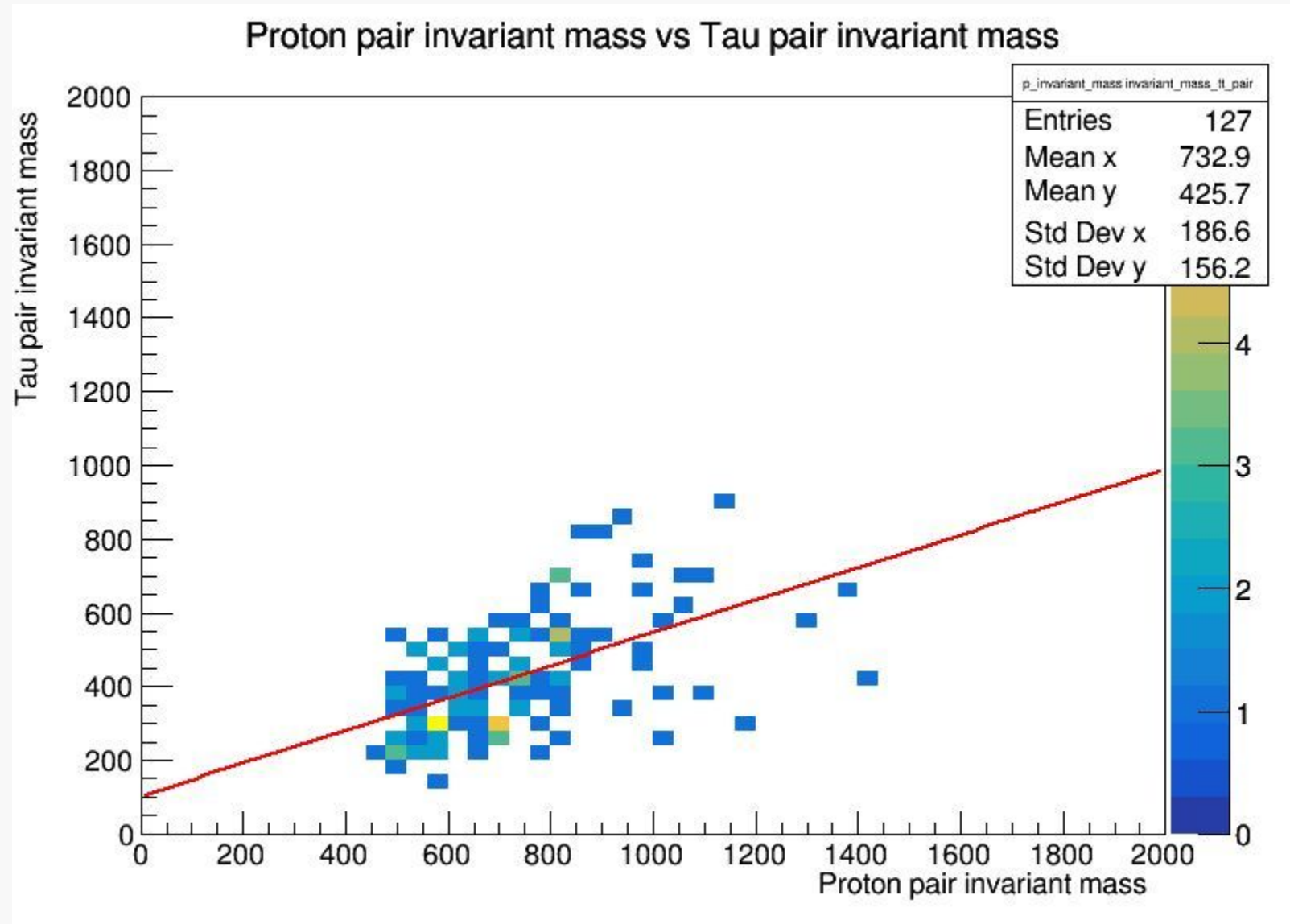
Change to formula that uses **TLorentzVector**

$$M_X^2 = s\xi_1\xi_2$$

Variable Correlations (signal)



Variable Correlations (signal)



Variable Correlations (signal)

Next Steps

- **Repeat for sample with more events**
 - **Add sample with QCD background**
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